

I. DERIVATION OF GENERAL RELATIVITY FROM THE SAT FRAMEWORK

The emergence of General Relativity (GR) from the SAT/Blockwave framework follows from the variational principle applied to the constrained tensorial action. The time-flow vector u^μ acts as a dynamically normalized timelike field whose fluctuations encode the spacetime metric curvature. The Einstein–Hilbert action thus arises as the continuum limit of the projected, constraint-enforced Lagrangian.

A. Constrained Tensor Action

From the normalization constraint,

$$\mathbf{C}_1 : \quad u^\mu u_\mu + 1 = 0, \quad (1)$$

we introduce a Lagrange multiplier $\lambda(x)$ enforcing the timelike nature of the flow. The gravitationally extended Lagrangian density becomes

$$\mathcal{L}_{\text{grav}} = \frac{1}{2\kappa} R + \lambda(u^\mu u_\mu + 1) + \alpha P_{\mu\rho} P_{\nu\sigma} (\nabla^\mu u^\nu)(\nabla^\rho u^\sigma), \quad (2)$$

where R is the Ricci scalar curvature and $P_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$ is the spatial projector onto the hypersurface orthogonal to u^μ .

B. Variation with Respect to the Metric

Variation of the total action,

$$S = \int d^4x \sqrt{-g} \mathcal{L}_{\text{grav}}, \quad (3)$$

with respect to the metric $g^{\mu\nu}$ yields

$$\delta S = \frac{1}{2\kappa} \int d^4x \sqrt{-g} (G_{\mu\nu} - \kappa T_{\mu\nu}) \delta g^{\mu\nu}, \quad (4)$$

where the Einstein tensor arises from the variation of R ,

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R, \quad (5)$$

and the effective stress–energy tensor receives contributions from the SAT fields and constraint terms:

$$T_{\mu\nu} = -2 \frac{\delta \mathcal{L}_{\text{SAT}}}{\delta g^{\mu\nu}} + g_{\mu\nu} \mathcal{L}_{\text{SAT}}. \quad (6)$$

Requiring $\delta S = 0$ for arbitrary variations $\delta g^{\mu\nu}$ immediately produces the Einstein field equations,

$$\boxed{G_{\mu\nu} = \kappa T_{\mu\nu}}. \quad (7)$$

C. Effective Stress–Energy Tensor

In the SAT sector, the projected kinetic term for u^μ [see Eq. (2)] contributes a stress–energy component:

$$T_{\mu\nu}^{(u)} = 2\alpha [(\nabla_\mu u_\rho)(\nabla_\nu u^\rho) - (\nabla_\rho u_\mu)(\nabla^\rho u_\nu)] - \alpha g_{\mu\nu} (\nabla_\rho u^\sigma)(\nabla^\rho u^\sigma) + 2\lambda u_\mu u_\nu. \quad (8)$$

This structure parallels the aether-type field models in Lorentz-violating gravity, but here $\lambda(x)$ dynamically enforces the SAT time-flow normalization, preserving diffeomorphism invariance.

D. Canonical Form and Hamiltonian Reduction

The Hamiltonian density associated with the gravitational SAT sector is obtained by Legendre transforming $\mathcal{L}_{\text{grav}}$ with respect to $\dot{g}_{ij}$ and $\dot{u}_i$. Using the projector formalism from the SATO–BLOCK Hamiltonian construction, the canonical Hamiltonian is expressed as

$$\mathcal{H}_{\text{GR+SAT}} = N\mathcal{H}_0 + N^i \mathcal{H}_i + \lambda(u^\mu u_\mu + 1), \quad (9)$$

where N and N^i are the ADM lapse and shift variables, and $\mathcal{H}_0, \mathcal{H}_i$ are the Hamiltonian and momentum constraints:

$$\mathcal{H}_0 = \frac{1}{\sqrt{h}} (\pi^{ij} \pi_{ij} - \frac{1}{2} \pi^2) - \sqrt{h} R^{(3)} + \mathcal{H}_u, \quad (10)$$

$$\mathcal{H}_i = -2\nabla_j \pi^j_i. \quad (11)$$

Here π^{ij} is the canonical momentum conjugate to h_{ij} , the spatial metric, and $\mathcal{H}_u$ represents the contribution of the projected u^μ sector from the SAT constraint.

E. Resulting Einstein–Hilbert Limit

In the limit $\alpha \rightarrow 0$ and with $\lambda(x)$ enforcing $u^\mu u_\mu = -1$ identically, the projector term vanishes and the action reduces to the standard Einstein–Hilbert form:

$$S_{\text{EH}} = \frac{1}{2\kappa} \int d^4x \sqrt{-g} R. \quad (12)$$

Thus, GR emerges as the low-energy, purely geometric limit of the SATO–BLOCK SAT theory, while the full framework retains higher-order corrections that encode discrete topological and filament contributions.